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Systematic review on model predictive control strategies applied to active thermal energy storage systems

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ABSTRACT

This paper presents a review of the application of model predictive control strategies to active thermal energy storage systems. To date, model predictive control has been used to manage such energy systems as heating, ventilation and air conditioning equipment or power generation plants. In all cases, the aim of the strategy has been to anticipate both production and consumption decisions to optimize the system performance, reducing the final energy cost. This ability of the strategy to forecast weather conditions and predict demand requirements in advance exceeds the performance of conventional control methods and made the strategy a very effective option to be coupled with active thermal energy storage systems. In this regard, this review paper presents the progress and results of the combination of these two technologies. The key contributions consist of a summary of the technical parameters employed, such as the prediction horizon length, the computational architecture approaches, the thermal energy storage material used and the influence of renewables in this kind of system. Additionally, the review summarises the latest enhancements to overcome computational issues and an analysis of the objective functions employed in each study, which were mainly focused to minimize the energy cost, the peak power and CO_2 emissions. A discussion about the strengths and weaknesses of this technology is provided, highlighting the difficulty of the strategy to operate with complicated physical models as the key limitation to overcome. Finally, some future guidelines to enhance the application of this strategy to control different sort of systems are detailed.

1. Introduction

The continuous increase in the average global temperature is reaching worrying levels that need to be tackled immediately [1]. The high consumption of fossil fuels to produce energy is contributing to this increase. Accordingly, within its sustainable development goals, the United Nations has defined the target of integrating climate change measures into national policies, strategies and planning [2]. Among the sources of such high energy consumption, the building and industrial sectors represent 30% and 25% of the total final energy consumption [3]. Therefore, searching for actions that contribute to reducing energy consumption from fossil fuels in both sectors is a key factor to finding a balance between maintaining current energy production and reducing greenhouse gases emissions, and which will contribute to moving towards a more sustainable society. One of the easiest actions to achieve such a balance is the deployment of renewable energies [4]. However, the intermittency in their production hampers the integration of renewable sources into buildings [5] and industries [6].

Therefore, thermal energy storage (TES) systems have appeared as good solutions for addressing the mismatch between renewable energy production and consumption, helping to increase its reliability. Additionally, this capacity can lead to significant benefits in both building [7] and industrial applications [8]. The basic principle of TES systems is the storage of energy from thermal equipment that can operate during peak renewable hours, producing energy surpluses to be further used by an energy sink. Thus, to make accurate TES operation decisions, a control strategy able to manage fluxes in the energy system can yield both energy and cost savings. Currently, this energy management is mostly carried out through traditional control strategies or others whose operation is based on predefined rules. However, the good performance

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of these strategies depends strongly on external variables. The more constant the external conditions are the higher the savings that can be achieved. Therefore, the solution to reach the full potential of TES systems lies in managing the storage system through a control strategy able to adapt to future conditions. In that sense, the integration of smart control techniques into the TES operation can lead to optimal management of the system, achieving significant improvements compared with TES systems without control or those managed through a conventional control strategy. The main variables that a smart control strategy needs to evaluate when managing the operation of a TES system are future energy demand profiles and the production hours of renewable energy sources.

All mentioned control abilities are gathered in model predictive control (MPC) strategies. Their operation is based on a dynamic model to forecast system behaviour, and optimize the actions to operate under the best sequence of decisions [9]. The potential of this way of working increases when the MPC strategy can decide to store energy during the high-performance hours of the system. In this scenario, the capacity of TES becomes essential, since it gives the MPC strategy the opportunity to take advantage of the energy produced either by renewable energy sources or during off-peak hours, and store this energy in the TES tank for future consumption. Therefore, the role of MPC in systems with TES is to optimize both charging and discharging of the TES, considering all the energy and environmental conditions of the system, to minimize both energy cost and consumption. Although the coupling of MPC strategies and TES systems is young, some researchers have started to apply MPC to TES systems in recent years, aiming to increase system performance [10]. To date, its application has mainly been focused on buildings, obtaining significant cost and energy savings and showing robustness against occupancy variations during operation [11]. In this framework, Yu et al. [12] assessed different control strategies to manage a TES system in the building framework and defined MPC as the best control method among those that manage the system according to the evolution and interaction of their own parameters. This working methodology allows the control strategy to reach the full potential of active and passive TES systems integrated in heating, cooling, and ventilation (HVAC) equipment. Likewise, Cole et al. [13] pointed out that to maximize the benefits of TES systems, it is necessary to include constrained controls, such as MPC, because this demonstrated its viability to amplify the cost effectiveness of TES in HVAC systems. In a wider study, Thieblemont et al. [14] reviewed the MPC application in buildings with TES systems, also highlighting the need to predict both loads and renewable energy production in order to increase TES efficiency. Thus, due to the requirement to predict future system operation conditions, the capacity of MPC to manage TES systems had a good acceptance for buildings, which allowed the integration of renewable energy sources within their energy operation to be increased [15]. Moreover, other than in HVAC equipment, the application of MPC was studied in energy distribution networks, such as district heating and cooling systems, demonstrating its potential to reduce both costs and greenhouse gases emissions [16].

Therefore, the aim of this study was to obtain a wide perspective of the achievements over the last decade in MPC strategies applied to controlling active TES systems. The main motivation was to identify the key aspects that contributed to enhancing the system performance, to analyse the level of maturity of MPC strategies, and to assess the impact of renewable energy sources on this kind of system to date, from a general point of view. To the authors knowledge, there are no works that have reviewed the state-of-the-art of this topic, detailing the operation features and technical settings of MPC strategies applied to both building and industrial energy systems with active TES. Therefore, this article provides a systematic review, whose main novelty lies in the description of the TES materials used in the literature, the evaluation of the influence of renewable energy sources and the analysis of the main MPC settings, detailing the computing approaches employed by researchers to date.

2. Methodology

The analysis proposed in this paper was developed using a systematic review methodology [17]. The first step was the query string definition that included all the documents that should be considered in the study. Table 1 defines the query employed in both the Web of Science and Elsevier Scopus databases to obtain a wider perspective of the studies carried out so far.

The study divided the analysis of the literature into two categories: heating and cooling systems, and power generation plants. For both areas, the study reviewed the main features related to the TES and the energy system itself and detailed the characteristics of the MPC used to control such systems.

Focusing on heating and cooling systems, the TES material was identified in all studies, the occupancy schedule of the building where the control system was used, and the type of study, considering that it could be either at the simulation or experimental level. Moreover, two additional aspects were highlighted from all the articles related to heating and cooling: the incorporation of time-of-use (ToU) tariff structures in the electricity supply and the level of integration of renewable energy. Regarding the power generation systems, the TES material used in all studies was specified, as well as the level of integration of renewable energy sources.

With respect to the analysis of the MPC features, the review details the same information for both heating and cooling systems, and power generation plants. First, the MPC settings of all documents were identified based on the following parameters:

- Prediction horizon: This details how far MPC looks ahead to optimize a determined objective function under constraints.
- Time step: This is the time between control updates and iterative horizon optimizations.

Second, the review classified all studies according to the design architecture of the MPC algorithms used. This organization was divided between centralized and distributed formulations:

- Centralized MPC: The system dynamics, their interactions and the disturbances involved in the system operation are managed in a single process executed in a unique host. This provides an easier way to develop MPC algorithms, although the computational time can

Table 1
Query string to search scientific documents related to TES and MPC.

QUERY string	
Web of Science	Scopus
(“thermal energy storage” OR “cold energy storage” OR “latent near/3 storage” OR “sensible near/3 storage” OR “thermochemical near/3 storage” OR "*sorption near/3 storage”) AND (“model predictive control”)	TITLE-ABS-KEY ((“thermal energy storage” OR “cold energy storage” OR “latent W/3 storage” OR “sensible W/3 storage” OR “thermochemical W/3 storage” OR "*sorption W/3 storage”) AND (“model predictive control”))

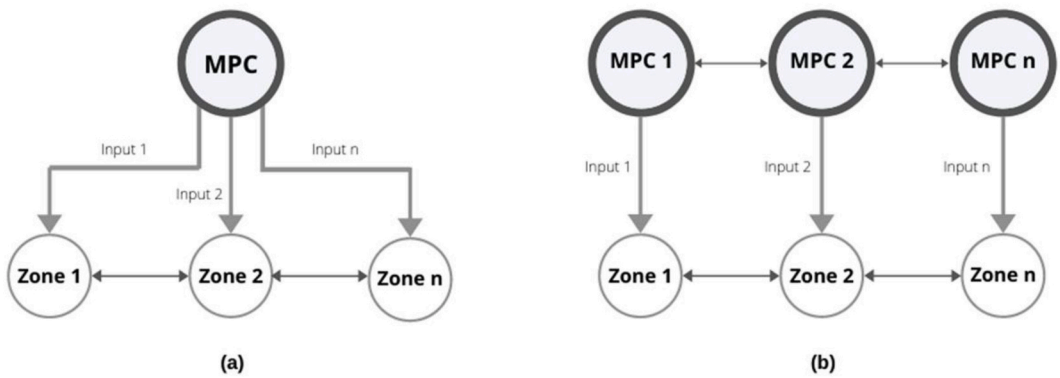

Fig. 1. Centralized MPC formulation (a) and distributed MPC formulation (b).

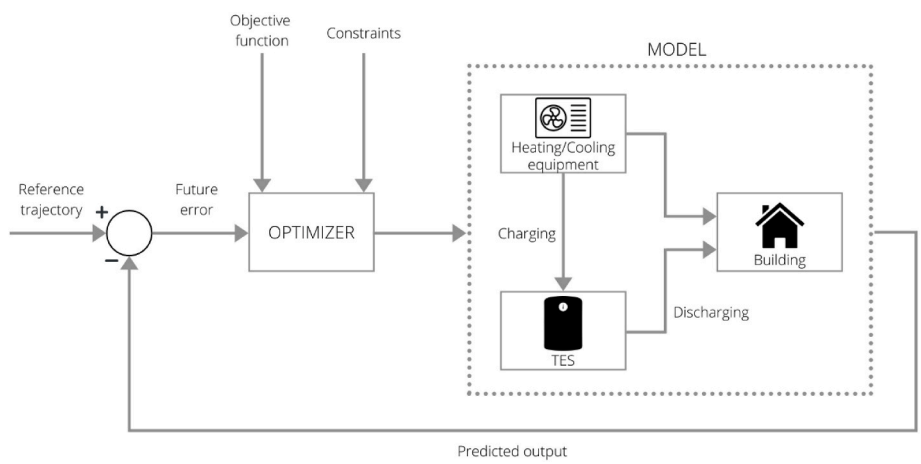

Fig. 2. Operation scheme of MPC in buildings with active TES.

become a drawback due to the large amount of data to be treated [18]. Fig. 1a presents a scheme of a centralized operation.

- Distributed MPC: It distributes different autonomous controllers throughout the control system operation, which are connected by a communication network. This architecture allows the simultaneous execution of the different pieces of the algorithm and provides high-speed resolutions. However, it is not easy to design as it has to manage the data dependencies between the different processes, as well as the distribution and gathering of partial results to conform a final result [19]. Fig. 1b shows a scheme of a distributed operation.

The third aspect considered in the analysis of the MPC strategies was to recognize the different approaches of the system:

- Deterministic MPC: The output of the model is fully determined by the parameter values and the initial conditions. From an MPC perspective, this approach is simple to solve, since there are no variations in the prediction.
- Stochastic MPC: This approach has some inherent randomness that is not easy to predict, and is more difficult to solve. However, its behaviour is closer to reality than deterministic models, due to the variability found in real life.

Moreover, the review identified some models that used specific methods to overcome the large computational effort that the use of MPC implies in certain energy systems. The characteristics of these approaches are summarised as follows:

- Hierarchical structures: These operate through several levels of action arranged in a treelike structure. The dynamics of the system are applied at each level, depending on the accuracy of its resolution, the speed of the resolution process, and the amount of data that must be managed.
- Hybrid controllers: A hybrid controller was considered one that contains more than one single control strategy, such as an MPC strategy with grey-box models or artificial neural networks (ANN), among others.

Finally, the objective function of all the documents reviewed was detailed, paying attention also to systems that were formulated with a multi-objective function. In case of multi-objective functions, MPC operated to minimize or maximize more than one system output.

3. Application of model predictive control in heating and cooling systems with active thermal energy storage

Although there are different control strategies in the literature able to manage both heating and cooling systems, MPC has shown the best performance in most studies that recently were carried out [20]. In comparison with traditional control strategies, MPC has the power to evaluate the outputs and corresponding inputs signals in advance. By doing so, MPC can avoid sudden changes in the control signal and the unexpected effects of delays in system response [21]. Additionally, the strengths of MPC compared with other smart control strategies, such as machine learning or ANN, are its smaller computation requirements and easier system modelling. These advantages encourage the implementation of an MPC strategy when controlling a system with active TES, since it is frequent to experience undesirable deviations. Moreover, controlling a TES tank in a heating or cooling system requires a control strategy with a good trade-off between computational time and accuracy in the

Table 2
Summary of MPC strategies applied to heating equipment with active TES.

Ref.	Year	TES material			Occupancy schedule				Type of study		ToU tariff		Renewable energy				MPC settings
		Water	PCM	N/A	Residential	Office	Service	N/A	Simulation	Experimental	Yes	No	PV panels	Solar collectors	Wind turbines	No	Prediction horizon
[29]	2014			•			•		•		•		•		•		1-h
[30]	2014	•			•				•			•		•			12-h
[31]	2015	•				•			•			•	•				48-h
[32]	2016	•			•				•		•					•	1,5,11,24-h
[33]	2017	•			•				•		•		•		•		96-h
[34]	2017			•	•					•	•		•	•			12-h
[35]	2018	•			•				•		•					•	24-h
[36]	2018		•					•	•	•		•				•	3 days
[37]	2018		•		•				•			•		•			24-h
[38]	2018		•		•	•	•		•		•			•			2,6,12,24-h
[39]	2018	•			•				•		•					•	24-h
[40]	2018	•			•				•		•		•				N/A
[41]	2019			•				•		•	•		•				5-6-h
[42]	2019	•			•				•		•					•	24-h
[43]	2019	•						•	•			•				•	24-h
[44]	2019	•			•				•		•					•	8-h
[45]	2019	•			•				•			•		•			1 year
[46]	2019	•			•				•			•	•				24-h
[47]	2019	•			•					•		•	•				24-h
[48]	2020	•			•				•		•		•				N/A
[49]	2020	•			•				•		•		•				24-h and 1-h
[50]	2020	•			•				•		•		•				24-h and 1-h
[51]	2020	•			•				•			•	•				24,36,48,96-h
[52]	2020	•			•				•		•					•	12-h
[53]	2020	•			•				•		•					•	12-h
[54]	2020	•			•					•	•			•			24-h
[55]	2020	•						•		•		•		•			50- min

results to maintain user comfort. Therefore, considering this framework, MPC was found to be an effective control strategy for dealing with all necessities.

To analyse the combination of MPC and active TES systems, the aim of this section is to provide an overview of the main technical characteristics in heating and cooling systems with active TES that are controlled by an MPC strategy. Although some studies included both heating and cooling supplies in the same system, either at the simulation level [22–26] or through experimental tests [27,28], most of them were assessed individually. In general, the way an MPC strategy operates in the building framework is depicted in the block scheme shown in Fig. 2. Considering a reference system trajectory and the feedback given by the model (predicted output), the strategy assesses the future error and this is used by the optimizer to evaluate the best consumption and production decisions of the system, to minimize/maximize a determined objective function. To take these decisions, the optimizer analyses the future system conditions based on the prediction horizon defined by the user and considering a set of different system constraints to operate within the pre-established system boundaries. After that, the optimizer gives the system model the energy fluxes that must be distributed between the heating/cooling equipment, the TES tank, and the building. The applications of heating and cooling systems with MPC and active TES are presented in Sections 3.1 and 3.2, respectively.

3.1. Heating systems

Two different types of heating systems with active TES controlled by an MPC strategy were identified in the literature: heating systems based on HVAC equipment (Section 3.1.1) and district heating networks (Section 3.1.2).

3.1.1. HVAC equipment

Due to the wide range of HVAC equipment used for heating, a general comparison of all these studies was carried out, considering the characteristics summarised in Table 2:

- The TES material used in the storage tank, either water or a phase change material (PCM). This election determined the complexity of the TES model, being the systems with PCM more difficult to tackle, due to the phase change modelling.
- The type of occupancy schedule that the heating system was working with. This could follow a residential, office, or service schedule, which affected the peak energy consumption periods, as well as the energy consumption profiles. The shape and homogeneity of the profile determined the affectation of the occupancy schedule upon the resolution process of the MPC strategy.
- The type of the study, differentiating between simulations and experimental tests.
- It was assessed whether ToU tariff structures were considered in the operation of each heating system.
- The level of renewable energy source integration in each study. This could include PV panels, solar collectors or wind turbines.
- The prediction horizon and the time step employed in all studies, with the purpose of defining the operational settings of each MPC strategy.
- Regarding the design of the MPC strategies, different aspects were considered to assess the main design approaches taken by the authors. First, the computer architecture of each MPC strategy was analysed. This could be centralized or distributed. The approach of the model was also evaluated, differentiating between deterministic and stochastic models. Moreover, the strategies that used hierarchical structures and hybrid models were pointed out.
- Finally, a review of the objective function used in all studies was carried out. The main goal was to identify which the main concern of the authors was when minimizing or maximizing one of the system

MPC settings	MPC approach						MPC objective function					
Time step	Centralized	Distributed	Deterministic	Stochastic	Hierarchical	Hybrid	Cost reduction	Peak reduction	CO_2 emissions reduction	Maximize energy selling	Maximize exergy in TES tank	Maximize TES stratification
2-min		•	•				•					
20-min	•		•				•					
30-min	•		•			•	•					
1-h	•		•				•	•				
1-h	•		•				•		•			
20-min	•		•			•	•					
1-h	•		•				•					
N/A	•		•		•	•	•					
1-h	•		•				•					
5,15,30,60-min	•		•				•					
1-h	•		•				•					
N/A		•	•				•					
5-10-min	•	•	•				•					
1-h	•		•				•					
1-h	•		•				•					
15-min and 1-h	•		•			•	•					
2 days	•		•								•	
15-min	•	•	•				•					
15-min	•		•				•					
N/A	•		•				•					
1-h and 5-min	•		•		•		•					
1-h and 5-min	•		•		•		•					
1 h	•		•							•		
15-min	•		•				•					
10-min	•		•			•	•					
1-h	•			•	•	•	•					
10- min	•		•		•	•						•

variables. Also, an analysis of the incidence of multi-objective functions of each MPC strategy was carried out.

As can be seen, water was the most commonly used TES material in this kind of systems. The number of articles that used PCMs was much smaller in comparison to those that used water, only three studies using a PCM as the TES material [36–38]. Among these studies, Papachristou et al. [36] used a cell with PCM that was modelled with a low-order thermal network model. Moreover, Serale et al. [37] managed a latent heat TES unit coupled to a solar thermal collector and a back-up electric heater. Similarly, Gholamibozanjani et al. [38] designed a system composed of a solar air collector, a heat exchanger filled with PCM, and a back-up heater. The three studies highlighted the complexity of modelling the behaviour of the temperature dependant specific heat capacity of the PCM. In contrast, the sensible working range of water could satisfy the heating requirements of all buildings, while it was easier to model [32,49,51,52].

Regarding the occupancy schedule, the majority of MPC strategies that managed a heating system were generally applied to residential occupancy schedules, except Ferrarini et al. [29], which used a service schedule, Li et al. [31] with an office schedule and Gholamibozanjani et al. [38] which compared the three occupancy schedules. The three works demonstrated the ability of MPC at the simulation level to manage the different consumption profiles and capacities of the strategies to optimize the interaction between the TES system and the building. However, there are no experimental tests in the literature that studied the effect of different occupancy patterns on MPC performance. In fact, only six articles tested MPC under experimental conditions. Among these, Khakimova et al. [34] tested an MPC strategy applied to managing the thermal subsystems of a small building. The results showed that the controller minimized the energy cost of the system without violating the established constraints. Another experimental study done by Rastegarpour et al. [41] validated their MPC strategy in a smart grid research laboratory. The results confirmed those obtained from previous simulations and showed the feasibility of online optimal control in real life cases. Other experimental studies were carried out by Kuboth et al. [47] and Finck et al. [54]. These tested the performance of two heating systems based on heat pumps, following the guidelines of an MPC strategy with the goal of reducing the final energy cost.

Focusing on the energy price in this kind of systems, it is crucial to assess the impact of ToU tariff structures. As the main heating source in the reviewed literature was based on heat pumps, the possibility of taking advantage of the off-peak periods was of special interest for reducing the final energy cost. In that sense, the incidence of ToU tariffs in the heating systems studied shown in Table 2 is notable, since more than half of them employed variable electricity pricing in their studies. For instance, Fitzpatrick et al. [48] evaluated the potential of MPC to manage the consumption decisions based on reducing the cost. They did so by analysing the daily operating schedule of the energy systems according to several ToU tariffs. The results demonstrated the potential of active TES systems to store energy during off-peak periods and consume it during peak consumption hours. More studies took this advantage into account [29,32–35,38–42,44,49,50,52–54], which encourages researchers to use ToU tariffs in future studies.

Additionally, to reduce the cost of the energy consumption even further, the integration of renewables within the operation of these heating systems is also a key aspect which some authors focused their attention on. Table 2 details the type of renewables that were used in the literature to mitigate the energy consumption coming from the grid. A combination of PV panels and wind turbines was used by Ferrarini et al. [29] and Baeten et al. [33]. These obtained significant cost reductions due to the ability of MPC to store the energy surpluses obtained by solar and wind productions in the TES tank. However, most of the studies only considered PV panels as their renewable energy source for the system, taking advantage of their easy inclusion into heating systems based on heat pumps. In that sense, Kuboth et al. demonstrated the potential of PV panels to reduce the final energy consumption of a heat pump at the simulation level [46] and in an experimental prototype [47]. Similarly,

Tarragona et al. [49,50] highlighted this potential in a heating system based on an air-to-water heat pump supplied by PV panels. Furthermore, other authors decided to use solar collectors to take advantage of solar energy. This was done by Jonin et al. [45] for a system with a seasonal TES. Serale et al. [37] and Gholamibozanjani et al. [38] also employed solar collectors in their systems with PCM as the TES storage material.

Once the main features of the systems have been explained from an energy point of view, it is also important to describe the control parameters and approaches employed by the authors in their MPC strategies. First, the settings of the MPC operation, such as the prediction horizon and the time step, were identified for all the studies analysed. Overall, it should be highlighted that most of the studies used a prediction horizon of between 12 h and 24 h, because this range was shown to be long enough to attain optimal cost reductions in the heating systems with TES studied. However, a specific study to assess the influence of such a prediction horizon when adding an electrical battery within the system operation was carried out by Langer and Volling [51]. The authors analysed the effect of different prediction horizons lengths (24, 36, 48 and 96 h), aiming to maximize the electrical energy selling coming from PV panels. The authors concluded that systems with the objective function of maximizing the profits and composed of both TES and electrical batteries needed longer horizons since the complexity of the system was higher. Besides, another study about the influence of the length of the prediction horizon was carried out by Sundström et al. [32]. In this case, the authors analysed the trade-off through a multi-objective MPC between minimizing the total power consumption cost and the peak consumption. The results showed that if the goal is to reduce peak power, the most important factor is to have a long prediction horizon since the energy management controller can plan and distribute the energy consumption better when this is the case. Another particular case was found in the literature in the study by Jonin et al. [45]. This used a prediction horizon of 1 year. As the TES was used as seasonal storage, the prediction to decide when to charge or discharge considered long horizons. However, in this case, the data time step employed to update the system status was 2 days, meaning that the global system operation followed a slow operating dynamic. In contrast, except for Ferrarini et al. [29] with a time step of 2 min, the range of the time-step value in the other studies was between 10 min and 1 h. Although it was shown that the higher the time step, the lower the accuracy in the final results, a time step of 1 h obtained a good trade-off between these two concepts [49]. Therefore, most of the simulations were performed with maximum time steps of 1 h. However, in the experimental studies the time step was reduced to react in advance to the variability derived from the real conditions [34,41,47,55].

Other factors to analyse in all MPC strategies were the programming architecture, which could be centralized or distributed, and the control approach, either deterministic or stochastic. In most of the studies, the MPC strategy was programmed using a centralized architecture and considering the studied system to be deterministic in nature. Few works employed distributed strategies in simulations [29,40,46] or experiments [41], and only one work considered a stochastic approach [54]. In all cases with a distributed MPC, the distributed architecture was employed to separate the decision variables and optimization solvers, and hence to overcome the computation limitations of the centralized architecture. Nevertheless, another solution for tackling the limitations of the centralized architecture was to employ hierarchical control structures. Through this option, authors divided the operation of the MPC into different levels to reduce the computational effort of the control system. In the literature, Tarragona et al. [49,50] created an MPC strategy with two levels of operation with a prediction horizon of 24 h at the high-level and 1 h at the low-level. Likewise, Finck et al. [54] created a hierarchical structure for a double-level economic MPC and Gil et al. [55] also designed a double-level controller to maximize the TES tank stratification. It is worth noting that Finck et al. [54] employed ANN to model the heat pump and a grey-box to model the TES tank, this being a hybrid system. In that sense, Gil et al. [55] also used a hybrid system in their hierarchical structure, which combined an MPC and a PID, the former being the strategy for the upper-level and the latter, the controller for the lower-level. More studies considered hybrid systems and added more than one method to their system modelling or operation [31,34,36,44,53].

Finally, focusing on the MPC objective function, the goal in almost all studies was to reduce the cost of system operation. However, two of these studies employed a multi-objective function and, other than minimizing the operating cost, they minimized the peak reduction in one case [32] and the CO_2 emissions in the other [33]. Additionally, other authors, such as Jonin et al. [45], Langer and Volling et al. [51], and Gil et al. [55] decided not to reduce the cost and focused the objective on maximizing respectively the exergy in the TES tank, the energy sales, and the TES tank stratification.

3.1.2. District heating networks

The incidence of MPC strategies and active TES was smaller in district heating networks than in heating systems based on HVAC equipment. However, there are some aspects that are important to consider. The first point to highlight is that no experimental works were found in the literature. Table 3 provides a summary of the district heating research carried out at the simulation level to date. As was done in Section 3.1.1, the summary provides the TES material, the occupancy schedule followed by the buildings where the district heating network operates, the renewable energy sources used in all studies and the MPC working characteristics. An interesting aspect that is worth pointing out from the district heating studies is that, except in the work by Eynard et al. [56] in 2012, all the documents were published in the last five years. This is a clear indicator that the coupling between MPC and TES is just starting to be explored in the framework of district heating networks. Moreover, another significant factor to highlight is that all the studies employed water as the TES material. As in heating studies based on HVAC equipment, the difficulty of modelling the PCM behaviour without increasing the computational effort of MPC was a strong limitation. Regarding the occupancy schedule under which the operation of district heating systems was analysed, residential was the most widely used. Only Eynard et al. [56] studied the behaviour of the MPC strategy under office and service occupancy schedules with active TES, and obtained significant cost reductions in all of these.

Concerning the analysis of the renewable energy sources employed in district heating networks, the influence of biomass should be highlighted. In fact, Eynard et al. [56] included a biomass boiler as a renewable energy source to analyse its performance in the three above-mentioned occupancy schedules. Labidi et al. [57,58] and Moustakidis et al. [59] also included a similar renewable energy source within their systems operation. Due to the high energy production needed by district heating networks, a biomass boiler was identified by these authors as a good option to satisfy such a requirement. Moreover, the use of biomass significantly contributes to reducing the reliance on fossil fuels and limiting carbon dioxide emissions [60]. The high-combustion efficiencies of modern biomass boilers led this renewable energy source to be particularly cost-effective in reducing the carbon footprint. Additionally, it is worth noting that biomass could reduce operational costs, since it is cheaper than conventional fossil fuel alternatives. The option to include PV panels and wind turbines instead of biomass was studied by Ruusu et al. [61], reaching a reduction in the cost of imported electricity of between 21 and 75% and noting both sources as interesting options for further research.

Focusing on the MPC settings employed in these studies, most used prediction horizons of 24 h and a time step of 1 h. However, an exception was identified in the above-mentioned system of Ruusu et al. [61], which reduced the time step to 6 min to assess better the variations induced by the double renewable energy source included in the system operation. On the other hand, Verrilli et al. [62,63] used a time step of 1 h, but reduced the prediction horizon to 12 h. In these cases, as the authors did not consider renewable energy sources, 12 h were enough to

Table 3
Summary of MPC strategies applied to district heating networks with active TES.

Ref.	Year	TES material	Occupancy schedule			Renewable energy			MPC settings			MPC approach					MPC objective function	
			Residential	Office	Service	N/A	PV panels	Wind turbines	Biomass	No	Prediction horizon	Time step	Centralized	Distributed	Deterministic	Stochastic	Hierarchical	Hybrid	Energy cost reduction	Maintenance cost reduction
[56]	2012	Water	•	•	•				•		4.5-h	5-min	•		•				•	
[57]	2015	Water	•						•		24-h	1-h	•			•		•	•	
[64]	2016	Water	•							•	24-h	1-h	•		•				•	
[63]	2016	Water	•							•	12-h	1-h	•			•			•	
[58]	2016	Water	•						•		24-h	1-h	•		•			•	•	
[62]	2017	Water	•							•	12-h	1-h	•		•				•	•
[61]	2019	Water	•				•	•			24-h	6-min	•		•		•		•	
[59]	2019	Water	•						•		24-h	1-h	•		•		•	•	•	
[65]	2020	Water				•				•	24-h	1-h	•		•				•	

optimize the operating cost [63] and minimize both the operation and maintenance costs of the district heating plant studied [62]. Additionally, another particularity was analysed by Verrilli et al. [63], because the authors designed an MPC strategy using a stochastic formulation to move the model towards a more realistic framework. The behaviour of this random approach was successfully managed by the designed MPC control tool, encouraging the authors to consider it in future studies. Finally, it is worth noting that the literature supplies two studies that used a hierarchical structure, and three with hybrid systems.

3.2. Cooling systems

Regarding the review of the literature related to cooling systems, the only studies found among all the publications reviewed applied MPC strategies to cooling systems based on HVAC equipment with active TES (Section 3.2.1). There were no works that considered this configuration for district cooling networks.

3.2.1. HVAC equipment

A summary of all MPC studies that contain cooling systems with active TES was done. Table 4 provides all the information extracted from these studies, detailing the same characteristics as in the heating systems shown in Section 3.1.1. First, it is worth noting that, unlike the heating systems previously studied, the cooling systems found in the literature used a higher variety of PCM materials to store their energy, these being mainly ice or other commercial PCMs. Regarding ice TES systems, Candanedo et al. [66] analysed the behaviour of a cooling system based on a chiller controlled by an MPC strategy to optimize the charging and discharging processes of an ice bank. The results highlighted the ability of MPC to optimize the operation of the ice bank. This was confirmed by Dehkordi and Candanedo [67] in a comparison of the same system operating under MPC and a rule-based control (RBC) strategy. Similarly, Beghi et al. [68,69] compared the ability of an MPC strategy to manage a cooling system with ice as TES material with conventional strategies. The results obtained through simulations showed that MPC was more effective in terms of energy efficiency and demand satisfaction than the other strategies analysed. Aiming to take advantage of the operational flexibility of the active TES, Pertzborn [70] and Cao et al. [71] also considered ice as a TES material to optimize the operation of their chiller systems, both located in service buildings. A step further was taken by Touretzky et al. [72], who also applied MPC in a service building, but in this case, the authors used a commercial PCM instead of ice. The capacity of the PCMs to store larger amounts of energy was important for operating in cooling systems located in service or office buildings, since cooling demands tend to be high. However, the authors highlighted the difficulty of solving the nonlinearities and tackling the binary decisions due to the nature of the PCM. To overcome this problem, in a further study, Touretzky and Baldea [73] proposed a hierarchical MPC strategy with a continuous reformulation of this scheduling problem. This approach showed that the strategy solved the problem in a short period of time, while accounting for the complexity and discontinuities associated with a detailed model of the PCM storage. In that sense, the same hierarchical approach for a TES system with a commercial PCM was developed by Wang et al. [74] and Pangborn et al. [75]. They also obtained good control system behaviour that coordinated the relatively slow phase change dynamics of the TES and the relatively fast temperature dynamics elsewhere.

Focusing on the occupancy schedule, almost all the cooling systems studied in the literature were in service and office buildings. Only Mendoza-Serrano and Chmielewski [76–78] carried out simulations for a residential occupancy schedule. Mostly, the tendency in these systems was to operate large cooling equipment with TES in buildings with cyclical demand profiles, as is the case of both office and service buildings. From a computational perspective, simple MPC configurations with a centralized architecture and deterministic approaches were enough to tackle the resolution of this type of systems, considering that

Table 4
Summary of MPC strategies applied to cooling equipment with active TES.

Ref.	Year	TES material				Occupancy schedule					Type of study		ToU tariff		Renewable energy	
		Water	Ice	Another PCM	N/A	Residential	Office	Service	Test room	N/A	Simulation	Experimental	Yes	No	PV panels	Solar collectors
[81]	2009	•						•			•		•			
[79]	2011	•						•				•	•			
[80]	2012	•						•				•	•			
[94]	2012	•					•	•			•		•			
[76]	2012				•	•					•		•			
[66]	2013		•					•			•		•			
[67]	2013		•					•			•		•			
[92]	2013	•						•			•		•			
[85]	2013	•					•				•		•			
[69]	2013		•					•			•		•			
[86]	2014	•							•		•		•			
[77]	2014				•	•					•		•			
[93]	2014	•						•			•		•			
[82]	2014	•					•				•		•			
[68]	2014		•					•			•		•			
[72]	2015			•				•			•		•			
[78]	2015				•	•					•		•			
[83]	2015	•					•				•		•		•	
[87]	2015		•				•				•		•			
[73]	2016			•					•		•		•			
[74]	2018			•					•		•			•		
[88]	2019	•						•			•			•		
[84]	2019	•					•				•			•		•
[70]	2019		•					•			•		•			
[71]	2019		•					•			•		•			
[89]	2019			•				•			•			•		
[95]	2019			•						•	•		•			
[90]	2020	•						•			•		•			
[75]	2020			•						•	•			•		
[91]	2020	•						•			•		•			

most of them operated without renewables. Then, regarding the type of experiments, all studies in the cooling framework were carried out through simulations, except the ones by Ma et al. [79,80]. In this latter case, the authors first validated the MPC strategy [79] previously designed and studied at the simulation level [81] in an university building where the goal was to minimize the energy cost. Later, a further step was taken by the authors, testing the design of a low-complexity MPC scheme in the same building. This had three objectives: to develop a simple switching non-linear model for the storage tank identified and validated by historical data, to integrate weather prediction into the MPC design to optimize the chiller operation, and to guarantee the robustness of MPC against uncertain building load demands [80].

Moreover, the use of ToU tariff structures and the contribution of renewable energy sources in the studies shown in Table 4 should be analysed in detail. While almost all the articles considered the ToU tariff structures in their operation, only four considered renewable energy sources within the cooling system operation. Therefore, it is clear that the authors pursued the peak load shifting potential that can be achieved with active TES systems that are controlled by MPC strategies. As most of the systems worked in office and commercial buildings with constant occupancy patterns, the capacity to adapt the charging and discharging decisions of the TES tank, together with the potential that MPC gave to forecast weather conditions and future changes on the dynamics of the cooling demand, were very useful for reducing the final energy cost. However, the low impact of renewable energy sources did not allow the consumption of energy from fossil fuels to be reduced. In fact, only Berkenkamp and Gwerder [82], Wang et al. [74], Zhao et al. [83], and Pintaldi et al. [84] employed renewable energy sources in their cooling systems. All the remaining works used only energy coming from the grid. This low ratio of renewable energy source integration led to part of the MPC potential being wasted, since one of the main capacities of MPC in systems with active TES is the improvement of the contribution of renewable energy systems.

Regarding the MPC settings, the prediction horizon in almost all the studies was 24 h. As mentioned, considering that many articles used ToU tariff structures, which have a day-ahead price profile, a 24-h prediction horizon was long enough to reduce the final cost. Longer horizons could not take more advantage of variability in electricity pricing. Similarly, the most commonly employed time step was 1 h, for the same reason, since the electricity price was set by the electricity companies every hour. However, to obtain higher accuracy in the final results and achieve an improved perspective of both cooling and TES dynamics, some authors reduced the time step in a range of between 10 min and 30 min [67–71,73,74,76,85–89]. Afterwards, as was done in the analysis of the heating systems, a summary of the main programming features of the MPC strategies was provided, identifying the MPC features detailed in Section 2. The computing architecture with most incidence in the literature was the centralized one. Only Kim [85] and Pertzborn [70] used a distributed MPC architecture, highlighting its fast computing ability compared to the centralized model. As in the distributed MPC control strategies applied to heating systems, the distributed models in cooling divided the computational load between multiple local models and optimizations, providing an effective global control policy for the entire operating regime. Focusing on the MPC approach followed by the MPC strategies studied, their evolution over the years is worth noting. While there was only one article that used a stochastic MPC between 2009 and 2015, the remaining studies being deterministic, from 2016 to 2020, five research papers that used a stochastic approach were published (almost 50% of the total publications). Therefore, the interest in applying models with random approaches increased and moved MPC towards more realistic operation conditions. Examples of this are the studies by Yu and Pavlak [90] and Lee et al. [91], which considered uncertainty in the occupant behaviour and weather of the following day. However, to manage the higher complexity driven by either stochastic models or more accurate physical models, it was necessary to employ hierarchical structures or hybrid models. For instance, the

Renewable energy		MPC settings		MPC approach						MPC objective function	
Free cooling	No	Prediction horizon	Time step	Centralized	Distributed	Deterministic	Stochastic	Hierarchical	Hybrid	Cost reduction	Other
	•	24-h	1-h	•		•				•	
	•	24-h	1-h	•		•				•	
	•	24-h	1-h	•		•				•	
	•	24-h	1-h	•		•				•	
	•	1,6,12-h	20-min	•		•				•	
	•	24-30-h	1-h	•		•				•	
	•	24,30,36,42-h	15-min	•		•				•	
	•	24-h	1-h	•		•		•	•	•	
	•	24-h	30-min		•	•				•	
	•	N/A	15-min	•		•				•	
	•	24-h	10-min	•		•		•	•	•	
	•	24-h	1-h	•		•				•	
	•	24-h	1-h	•		•		•	•	•	
•		24-h	1-h	•		•			•	•	
	•	N/A	15-min	•		•				•	
	•	24-h	1-h	•		•				•	
	•	24-h and 2-h	1-h	•		•				•	
		24-h	1-h	•		•				•	
	•	24-h	30-min	•			•			•	•
	•	24-h	10-min	•		•		•	•	•	
•		24-h	1-min and 20-min	•			•	•		•	
	•	24-h	30-min	•		•				•	
		18-h	1-h	•		•			•	•	
	•	48-h	15-min		•	•			•	•	
	•	24-h	30-min	•			•			•	•
	•	15-min	1-sec	•			•		•	•	
	•	12-h	1-h	•		•				•	
	•	N/A	1-h	•		•	•			•	
	•	N/A	N/A	•		•		•		•	
	•	24-h	1-h	•			•		•	•	

aforementioned article by Lee et al. [91] used an ANN model to predict the future behaviour of the building energy systems and tackle the variations in occupancy. Another example was designed by Pintaldi et al. [84]. This coupled MPC and genetic algorithms to minimize the back-up energy consumption while satisfying the cooling demand in a solar cooling system. Other studies without stochastic approaches included other strategies. This was the case of Deng et al. [92,93], who included dynamic programming models within the MPC operation.

Regarding the objective function, the goal in all the studies was to reduce the final operating cost. This tendency towards reducing the final cost reinforces the capacity of MPC to achieve peak load shifting and its high effectiveness in systems with active TES.

4. Application of model predictive control in power generation plants with active thermal energy storage

The influence of MPC on power generation plants with active TES is still in the early stages of deployment. This is illustrated by two considerations: only CSP and CHP plants integrated MPC control strategies with TES into their system operation and all publications were framed in recent years. The main features of MPC strategies applied respectively to CSP and CHP plants with active TES systems are explained in Sections 4.1 and 4.2.

4.1. Concentrating solar power plants

All studies in CSP plants are summarised in Table 5, where the main MPC technical features, as well as the TES material employed in all systems, are detailed. As can be seen, molten salt was the only material employed to store the thermal surpluses produced by the CSP plant.

Afterwards, paying attention to the MPC settings, a common prediction horizon between 24 and 48 h was used with a time step of 1 h [96–99,101]. In all these studies, the prediction horizon was based on a sliding window formulation. This method was used to consider the CSP plant generation schedule for the next hours and the following day. Then, using the sliding window prediction horizon, short and long intervals of predictions could be considered, depending on the knowledge requirements of the system.

Regarding other approaches found in the literature, Toub et al. [100,102] designed a micro-CSP plant to operate in a building. This system controlled by an MPC strategy outperformed the results obtained with traditional control strategies [100] and it was demonstrated to be a good alternative to solar PV panels [102]. Furthermore, Liang and Li [103] used a shorter prediction horizon and time step, because one of the aims of the work was to protect the turbine against possible damage. Therefore, it was necessary to use short data updating to control the evolution of the steam parameters more accurately.

Focusing on the MPC characteristics, all the studies used a centralized computing architecture and almost all of them, a deterministic approach. Only Toub et al. [102] simulated an MPC strategy based on a stochastic approach to consider the variability in the solar radiation, so improving the reliability of the input data. In addition, no strategies with a hierarchical structure or hybrid controllers were found in the literature.

Regarding the objective function of these studies, it is worth noting that minimizing the operating cost was not the only predominant goal. Considering that CSP plants can sell electricity to the grid, maximizing revenues from these sales as well as minimizing the penalties for changes in the prediction of the electricity production were two important aspects that were taken into account. Additionally, another common objective function in these works was to protect the turbine against possible malfunctioning or damage due to repeated start-ups.

4.2. Combined heat and power plants

So far, the use of MPC in CHP plants with active TES was very small,

Table 5
Summary of MPC strategies applied to CSP plants with active TES.

Ref.	Year	TES material		MPC settings		MPC approach						MPC objective function			
		Molten salt	N/A	Prediction horizon	Time step	Centralized	Distributed	Deterministic	Stochastic	Hierarchical	Hybrid	Minimize the operation cost	Maximize revenues	Minimize penalties for changes in electricity	Minimize turbine startup
[96]	2016	•		24,48-h	1-h	•		•				•			
[97]	2017	•		24,48-h	1-h	•		•					•		
[98]	2018	•		24,48-h	1-h	•		•						•	•
[99]	2018	•		24,48-h	1-h	•		•						•	•
[100]	2018		•	12-h	30-min	•		•				•			
[101]	2019	•		24,48-h	1-h	•		•						•	•
[102]	2019		•	24-h	30-min	•			•			•			
[103]	2019	•		500-sec	1-sec	•		•				•			•

Table 6
Summary of MPC strategies applied to CHP plants with active TES.

Ref.	Year	TES material		Renewable energy			MPC settings		MPC approach					MPC objective function	
		Water	N/A	PV panels	Wind turbines	No	Prediction horizon	Time step	Centralized	Distributed	Deterministic	Stochastic	Hierarchical	Minimize power imbalance	Minimize operation costs
[104]	2014	•				•	5-min	1-min	•	•	•			•	
[105]	2015	•				•	100-min	10-min	•		•				•
[106]	2016	•				•	1,12,24-h	1-h	•		•		•		•
[107]	2017		•		•		N/A	5-min	•		•				•
[108]	2019		•	•	•		N/A	N/A	•			•			•

as only five studies were found in the literature. Therefore, assessing the potential of MPC in these power generation systems is difficult as more works are needed to obtain broader conclusions about the system behaviour. Table 6 presents a summary of the main characteristics identified in these systems. Until now, the only material that was used for TES purposes was water [104–106]. In terms of renewable energy sources, only two studies considered these in CHP operations. Wu et al. [107] employed wind turbines in the CHP plant, aiming to produce heat for a municipal network and electricity for a provincial region, improving the overall economic efficiency of the plant with the decisions taken by MPC. A step further was taken by Zhang et al. [108], who included both wind turbines and PV panels. In this case, the switch-on/switch-off of the CHP plant was controlled to reduce the generation cost, taking advantage of both renewable energy sources.

Regarding the MPC settings, short prediction horizons and time steps were used in this kind of power generation plant, since it was very important to capture the changes during electricity and thermal production. An exception was the study performed by Wakui et al. [106], where a CHP plant was simplified to analyse its long-term dynamics. Thus, a linear approximation was carried out to study the plant output trajectories in longer prediction horizons.

Centralized computer architecture was used in all studies. However, Larsen et al. [104] also analysed the performance of a distributed architecture and compared it with a centralized one. The results from this comparison showed that the distributed architecture can be solved faster even while increasing the number of system variables, since the number of optimization variables per node was constant. With regard to the nature of the system, only Zhang et al. [108] considered a stochastic MPC, the deterministic version being the one used in the other studies. Likewise, only one work considered a hierarchical structure. Finally, regarding the objective functions used, almost all the studies aimed to minimize the operation system cost, except Larsen et al. [104], where the goal was to balance the local heat demand and supply.

5. Discussion

This review of the state-of-the-art details the main characteristics of MPC strategies applied to systems with active TES. Although the query string used to obtain the literature on the topic found many papers, some works were discarded, as they did not use MPC directly. In fact, in most of these articles, MPC was highlighted as a promising strategy for improving system operations. Indeed, the impact of MPC on energy systems with active TES is drawing attention from some researchers as a leading strategy for improving the performance of energy systems.

Overall, the performance of MPC in systems with active TES shown in the present review demonstrated robustness in many types of systems and reliability in their operation. All the studies used the combination of both MPC and active TES to reduce the mismatch between renewable energy production and peak consumption periods. By doing so, the systems could take advantage of higher renewable production ratios, while the influence of fossil fuels decreased. To have a better understanding of how the coupling between MPC and active TES systems to enhance the use renewable energy sources works, Fig. 3 shows a scheme of the energy flux breakdown in this kind of system. As it can be seen, MPC manages two types of energy production: constant and variable. Then, depending on the energy requirements, the strategy can decide to supply the energy produced by renewables directly to the building/industry or can store this for future consumption. The energy stored in the TES tank can be employed by the system when the MPC strategy detects that there is not enough renewable energy production to satisfy the demand requirements.

Several studies pursued the goal of increasing the share of renewable energy and avoiding the consumption of fossil fuels. However, most works developed powerful control tools at the simulation level, but only a few validated their behaviour in testing environments. From a global point of view, an accepted method to assess the level of maturity of this kind of systems is the technology readiness level (TRL) [109]. Most of the studies in this review developed powerful control tools at the simulation level, reaching level four on the TRL scale but only a few of them validated their behaviour in testing environments (TRL 5). Accordingly, the technology development of MPC in active TES was significantly achieved in the literature. However, further steps to test its behaviour in relevant environments should be implemented to promote the deployment of the control tool and increase its maturity. In that sense, this review identified the strengths and weakness of MPC control strategies in active TES systems as a way to contribute to future progress and to raise the TRL level of those systems:

- Strengths: The robustness to disturbances shown by MPC in different energy systems with active TES and its ability to adapt its control actions to different energy consumption profiles were identified as the most notable strengths of MPC in those systems. The integration of MPC was mostly in heating and cooling systems, where the effective interaction between MPC and heat pumps promotes the use of this control strategy. In fact, the capacity of TES to store energy and the predictive ability of MPC led to advantage being taken of the efficient electric performance of heat pumps, while reducing the operating costs and allowed an increase in the use of renewable energy sources. In computational terms, although the models were simple, MPC was demonstrated to be a cost-effective control tool which behaved well in systems with multi-objective functions. Moreover, the possibility of introducing constraints naturally made the system operation more reliable, since the energy thresholds could be maintained within the established bounds.

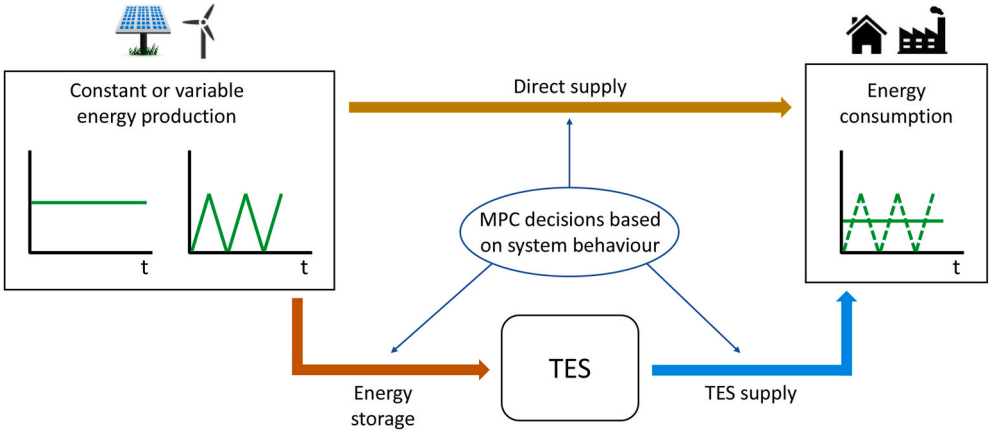

Fig. 3. Scheme of the renewable energy management performed by MPC strategies in systems with active TES.

- Weaknesses: The main drawback identified in this review was that the anatomy of MPC does not allow the strategy to operate with complicated physical models, due to the high computational complexity of the algorithm. The simple PCM models employed in the literature, as well as in most of the studies considered fully mixed TES models instead of stratified tanks illustrated this limitation. Additionally, this weakness affected the number of experimental studies carried out to date, since the greatest difficulty is to design a standalone MPC strategy able to solve the optimization process iterations in real time.

So far, the MPC models studied in the literature were lightweight in terms of modelling. The anatomy of MPC does not allow operating with physically complicated models, the computational burden being its main limitation. This was demonstrated in the small number of papers that included PCM as storage material, since its phase change range was difficult to solve in MPC strategies.

6. Conclusions and future work

This paper presents a review of active TES systems managed by MPC strategies. The articles were categorised into heating and cooling systems or power generation plants. Then, a comparison in terms of TES materials, demand profile working conditions, renewable energy sources integration, and MPC characteristics was carried out. It is worth remarking that most of the studies were carried out on the simulation level, considering ideal predictions and centralized computing architectures.

Regarding TES systems, water was used as the storage medium in most of the heating studies for two reasons: it was easier to model and store energy through sensible heat satisfied the energy requirements of the studied buildings. On the other hand, as most of the cooling systems studied operated in service and office buildings, which needed higher TES capacities, ice and commercial PCMs were also employed. Furthermore, in cooling systems, many studies considered ToU tariff structures in their operation. However, the level of renewable energy source integration was significantly low. Therefore, greater interest in pursuing the peak load shifting ability of MPC was perceived from this tendency.

Focusing on power generation systems, there were fewer research studies of MPC strategies and active TES in comparison with those with heating and cooling equipment. All the studies were from the last six years with a notable tendency to increase, meaning that the implementation of MPC is in a fledgling state in this system framework. Molten salts and water were the materials most widely used respectively in CSP and CHP plants.

Although many studies were conducted in the literature, further work is needed to fully explore this research field. Most of the MPC strategies were designed around a centralized computing architecture, presenting problems with the resolution process. Thus, more effort should be focused on designing distributed MPC strategies able to split this computing effort among a network of faster MPC controllers. Moreover, the capacity of hybrid systems, which employ more than one control strategy to improve the accuracy of the physical elements modelling, should be pointed out.

Regarding the potential of MPC, its capacity to reduce the final energy cost of an energy system was explored and demonstrated. However, there is a lack of knowledge about the power of the strategy to mitigate CO_2 emissions or optimize the operation of smart grids, which are emerging as the future world electricity transmission. Moreover, the implementation of more stochastic approaches that include variances in prediction can contribute to improving the reliability of the system.

All these future work suggestions require a big effort to couple two different technological areas: energy and computer science. The complexity of mixing energy systems, software and sensors become the main drawback so far. However, the emergence of The Internet of Things and the capacity of the new microcontrollers to exchange data from physical elements to control elements are the first steps to validate MPC control tools in real prototypes and to increase the potential of the strategy in energy systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] International Energy Agency. World energy outlook 2019. Paris. 2019.
[2] United Nations. The 2030 agenda for sustainable development. 2016.
[3] Global IEA. Status report for buildings and construction 2019. Paris. 2019.
[4] Panwar NL, Kaushik SC, Kothari S. Role of renewable energy sources in environmental protection: a review. Renew Sustain Energy Rev 2011;15: 1513–24. https://doi.org/10.1016/j.rser.2010.11.037.
[5] Ürge-Vorsatz D, Cabeza LF, Serrano S, Barreneche C, Petrichenko K. Heating and cooling energy trends and drivers in buildings. Renew Sustain Energy Rev 2015; 41:85–98. https://doi.org/10.1016/j.rser.2014.08.039.
[6] Raj NT, Iniyan S, Goic R. A review of renewable energy based cogeneration technologies. Renew Sustain Energy Rev 2011;15:3640–8. https://doi.org/10.1016/j.rser.2011.06.003.
[7] Parameshwaran R, Kalaiselvam S, Harikrishnan S, Elayaperumal A. Sustainable thermal energy storage technologies for buildings: a review. Renew Sustain Energy Rev 2012;16:2394–433. https://doi.org/10.1016/j.rser.2012.01.058.
[8] Pelay U, Luo L, Fan Y, Stitou D, Rood M. Thermal energy storage systems for concentrated solar power plants. Renew Sustain Energy Rev 2017;79:82–100. https://doi.org/10.1016/j.rser.2017.03.139.
[9] Rawlings JB, Mayne DQ. Model predictive control: theory and design (2009) + postface (2012). Automatica 2014;25:533.
[10] Tarragona J, de Gracia A, Cabeza LF. Bibliometric analysis of smart control applications in thermal energy storage systems. A model predictive control approach. J Energy Storage 2020;32:101704.
[11] Naylor S, Gillott M, Lau T. A review of occupant-centric building control strategies to reduce building energy use. Renew Sustain Energy Rev 2018;96: 1–10. https://doi.org/10.1016/j.rser.2018.07.019.
[12] Yu Z, Huang G, Haghighat F, Li H, Zhang G. Control strategies for integration of thermal energy storage into buildings: state-of-the-art review. Energy Build 2015; 106:203–15. https://doi.org/10.1016/j.enbuild.2015.05.038.
[13] Cole WJ, Powell KM, Edgar TF. Optimization and advanced control of thermal energy storage systems. Rev Chem Eng 2012;28:81–99. https://doi.org/10.1515/revce-2011-0018.
[14] Thieblemont H, Haghighat F, Ooka R, Moreau A. Predictive control strategies based on weather forecast in buildings with energy storage system: a review of the state-of-the art. Energy Build 2017;153:485–500. https://doi.org/10.1016/j.enbuild.2017.08.010.
[15] Sultana WR, Sahoo SK, Sukchai S, Yamuna S, Venkatesh D. A review on state of art development of model predictive control for renewable energy applications. Renew Sustain Energy Rev 2017;76:391–406. https://doi.org/10.1016/j.rser.2017.03.058.
[16] Zabala L, Febres J, Sterling R, López S, Keane M. Virtual testbed for model predictive control development in district cooling systems. Renew Sustain Energy Rev 2020;129:109920. https://doi.org/10.1016/j.rser.2020.109920.
[17] Lamb WF, Callaghan MW, Creutzig F, Khosla R, Minx JC. The literature landscape on 1.5 °C climate change and cities. Curr Opin Environ Sustain 2018;30:26–34. https://doi.org/10.1016/j.cosust.2018.02.008.

[18] Seal S, Boulet B, Dehkordi VR. Centralized model predictive control strategy for thermal comfort and residential energy management. Energy 2020;212:118456. https://doi.org/10.1016/j.energy.2020.118456.
[19] Venkat AN, Hiskens IA, Rawlings JB, Wright SJ. Distributed MPC strategies with application to power system automatic generation control. IEEE Trans Contr Syst Technol 2008;16:1192–206. https://doi.org/10.1109/TCST.2008.919414.
[20] Afram A, Janabi-Sharifi F. Theory and applications of HVAC control systems – a review of model predictive control (MPC). Build Environ 2014;72:343–55. https://doi.org/10.1016/j.buildenv.2013.11.016.
[21] Prívara S, Široký J, Ferkl L, Cigler J. Model predictive control of a building heating system: the first experience. Energy Build 2011;43:564–72. https://doi.org/10.1016/j.enbuild.2010.10.022.
[22] Dullinger C, Struckl W, Kozek M. A general approach for mixed-integer predictive control of HVAC systems using MILP. Appl Therm Eng 2018;128:1646–59. https://doi.org/10.1016/j.applthermaleng.2017.09.104.
[23] Zhang Y, Wang R, Zhang T, Liu Y, Guo B. Model predictive control-based operation management for a residential microgrid with considering forecast uncertainties and demand response strategies. IET Gener, Transm Distrib 2016; 10:2367–78. https://doi.org/10.1049/iet-gtd.2015.1127.
[24] Bartolucci L, Cordiner S, Mulone V, Santarelli M. Ancillary services provided by hybrid residential renewable energy systems through thermal and electrochemical storage systems. Energies 2019;12:2429. https://doi.org/10.3390/en12122429.
[25] Moser A, Muschick D, Gölles M, Nageler P, Schranzhofer H, Mach T. A MILP-based modular energy management system for urban multi-energy systems : performance and sensitivity analysis. Appl Energy 2020;261:114342. https://doi.org/10.1016/j.apenergy.2019.114342.
[26] Lee Z, Gupta K, Kircher KJ, Zhang KM. Mixed-integer model predictive control of variable-speed heat pumps. Energy Build 2019;198:75–83. https://doi.org/10.1016/j.enbuild.2019.05.060.
[27] Fiorentini M, Cooper P, Ma Z, Robinson DA. Hybrid model predictive control of a residential HVAC system with PVT energy generation and PCM thermal storage. Energy Procedia 2015;83:21–30. https://doi.org/10.1016/j.egypro.2015.12.192.
[28] Pean T, Costa-Castello R, Fuentes E, Salom J. Experimental testing of variable speed heat pump control strategies for enhancing energy flexibility in buildings. IEEE Access 2019;7:37071–87. https://doi.org/10.1109/ACCESS.2019.2903084.
[29] Ferrarini L, Mantovani G, Costanzo GT. A distributed model predictive control approach for the integration of flexible loads, storage and renewables. IEEE Int Symp Ind Electron 2014. https://doi.org/10.1109/ISIE.2014.6864871. 1700–5.
[30] Felix M, Lerch W, Heinz A, Goertler G, Schranzhofer H. A novel linear predictive control approach for auxiliary energy supply to a solar thermal. Combistorage 2014;101:203–19. https://doi.org/10.1016/j.solener.2013.12.015.
[31] Li S, Joe J, Hu J, Karava P. System identification and model-predictive control of office buildings with integrated photovoltaic-thermal collectors, radiant floor heating and active thermal storage. Sol Energy 2015;113:139–57. https://doi.org/10.1016/j.solener.2014.11.024.
[32] Sundström C, Jung D, Blom A. Analysis of optimal energy management in smart homes using MPC. In: 2016 eur. Control conf., aalborg, Denmark; 2016. p. 2066–71.
[33] Baeten B, Rogiers F, Helsen L. Reduction of heat pump induced peak electricity use and required generation capacity through thermal energy storage and demand response. Appl Energy 2017;195:184–95. https://doi.org/10.1016/j.apenergy.2017.03.055.
[34] Khakimova A, Kusatayeva A, Shamshimova A, Sharipova D, Bemporad A, Familiant Y, et al. Optimal energy management of a small-size building via hybrid model predictive control ☆. Energy Build 2017;140:1–8. https://doi.org/10.1016/j.enbuild.2017.01.045.
[35] D'Ettorre F, De Rosa M, Conti P, Schito E, Testi D, Finn DP. Economic assessment of flexibility offered by an optimally controlled hybrid heat pump generator: a case study for residential building. Energy Procedia 2018;148:1222–9. https://doi.org/10.1016/j.egypro.2018.08.008.
[36] Papachristou AC, Vallianos CA, Dermardiros V, Athienitis AK, Candanedo JA. A numerical and experimental study of a simple model-based predictive control strategy in a perimeter zone with phase change material. Sci Technol Built Environ 2018;24:933–44. https://doi.org/10.1080/23744731.2018.1438011.
[37] Serale G, Fiorentini M, Capozzoli A, Cooper P, Perino M. Formulation of a model predictive control algorithm to enhance the performance of a latent heat solar thermal system. Energy Convers Manag 2018;173:438–49. https://doi.org/10.1016/j.enconman.2018.07.099.
[38] Gholamibozanjani G, Tarragona J, Gracia A de, Fernández C, Cabeza LF, Farid MM. Model predictive control strategy applied to different types of building for space heating. Appl Energy 2018;231:959–71. https://doi.org/10.1016/j.apenergy.2018.09.181.
[39] Felten B, Weber C. The value(s) of flexible heat pumps – assessment of technical and economic conditions. Appl Energy 2018;228:1292–319. https://doi.org/10.1016/j.apenergy.2018.06.031.
[40] Rastegarpour S, Ghaemi M, Ferrarmi L. A predictive control strategy for energy management in buildings with radiant floors and thermal storage. SICE ISCS 2018;2018– Janua:67–73. https://doi.org/10.23919/SICEISCS.2018.8330158. 2018 - 2018 SICE int. Symp. Control syst.
[41] Rastegarpour S, Ferrarini L, Palaiogiannis F. A distributed predictive control of energy resources in radiant floor buildings. J Dyn Syst Meas Control Trans ASME 2019;141:1–63. https://doi.org/10.1115/1.4043935.
[42] D'Ettorre F, De Rosa M, Conti P, Testi D, Finn D. Mapping the energy flexibility potential of single buildings equipped with optimally-controlled heat pump, gas boilers and thermal storage. Sustain Cities Soc 2019;50:101689. https://doi.org/10.1016/j.scs.2019.101689.
[43] D'Ettorre F, Conti P, Schito E, Testi D. Model predictive control of a hybrid heat pump system and impact of the prediction horizon on cost-saving potential and optimal storage capacity. Appl Therm Eng 2019;148:524–35. https://doi.org/10.1016/j.applthermaleng.2018.11.063.
[44] Seal S, Dehkordi VR, Boulet B, Seal S. Coordination of radiant floor and baseboard heating systems : sequential and simultaneous MPC schemes Coordination of radiant floor and baseboard heating systems : sequential and simultaneous MPC schemes. Sci Technol Built Environ 2019:1–18. https://doi.org/10.1080/23744731.2019.1626166. 0.
[45] Jonin M, Khosravi M, Eichler A, Villasmil W, Schuetz P, Jones CN, et al. Exergy-based model predictive control for design and control of a seasonal thermal energy storage system. J Phys Conf Ser 2019;1343:012066. https://doi.org/10.1088/1742-6596/1343/1/012066.
[46] Kuboth S, Heberle F, König-haagen A, Brüggemann D. Economic model predictive control of combined thermal and electric residential building energy systems. Appl Energy 2019;240:372–85. https://doi.org/10.1016/j.apenergy.2019.01.097.
[47] Kuboth S, Heberle F, Weith T, Welzl M, König-Haagen A, Brüggemann D. Experimental short-term investigation of model predictive heat pump control in residential buildings. Energy Build 2019;204:109444. https://doi.org/10.1016/j.enbuild.2019.109444.
[48] Fitzpatrick P, D'Ettorre F, De Rosa M, Yadack M, Eicker U, Finn DP. Influence of electricity prices on energy flexibility of integrated hybrid heat pump and thermal storage systems in a residential building. Energy Build 2020;223:110142. https://doi.org/10.1016/j.enbuild.2020.110142.
[49] Tarragona J, Fernández C, de Gracia A. Model predictive control applied to a heating system with PV panels and thermal energy storage. Energy 2020;197: 117229. https://doi.org/10.1016/j.energy.2020.117229.
[50] Tarragona J, Fernández C, Cabeza LF, de Gracia A. Economic evaluation of a hybrid heating system in different climate zones based on model predictive control. Energy Convers Manag 2020;221:113205. https://doi.org/10.1016/j.enconman.2020.113205.
[51] Langer L, Volling T. An optimal home energy management system for modulating heat pumps and photovoltaic systems. Appl Energy 2020;278:115661. https://doi.org/10.1016/j.apenergy.2020.115661.
[52] Rastegarpour S, Gros S, Ferrarini L. MPC approaches for modulating air-to-water heat pumps in radiant-floor buildings. Contr Eng Pract 2020;95:104209. https://doi.org/10.1016/j.conengprac.2019.104209.
[53] Bechtel S, Rafii-Tabrizi S, Scholzen F, Hadji-Minaglou J-R, Maas S. Influence of thermal energy storage and heat pump parametrization for demand-side-management in a nearly-zero-energy-building using model predictive control. Energy Build 2020;226:110364. https://doi.org/10.1016/j.enbuild.2020.110364.
[54] Finck C, Li R, Zeiler W. Optimal control of demand flexibility under real-time pricing for heating systems in buildings: a real-life demonstration. Appl Energy 2020;263:114671. https://doi.org/10.1016/j.apenergy.2020.114671.
[55] Gil JD, Roca L, Zaragoza G, Normey-Rico JE, Berenguel M. Hierarchical control for the start-up procedure of solar thermal fields with direct storage. Contr Eng Pract 2020;95:104254. https://doi.org/10.1016/j.conengprac.2019.104254.
[56] Eynard J, Grieu S, Polit M. Predictive control and thermal energy storage for optimizing a multi-energy district boiler. J Process Contr 2012;22:1246–55. https://doi.org/10.1016/j.jprocont.2012.05.011.
[57] Labidi M, Eynard J, Faugeroux O, Grieu S. Predictive control of thermal storage systems designed for multi-energy district boilers: a case study in France. In: Proc. ISES sol. World congr. 2015, freiburg, Germany: international solar energy society; 2016. p. 1–11. https://doi.org/10.18086/swc.2015.02.09.
[58] Labidi M, Eynard J, Faugeroux O, Grieu S. A new strategy based on power demand forecasting to the management of multi-energy district boilers equipped with hot water tanks. Appl Therm Eng 2017;113:1366–80. https://doi.org/10.1016/j.applthermaleng.2016.11.151.
[59] Moustakidis S, Meintanis I, Halikias G, Karcanias N. An innovative control framework for district heating systems: conceptualisation and preliminary results. Resources 2019;8:27. https://doi.org/10.3390/resources8010027.
[60] Kitzing L, Mitchell C, Morthorst PE. Renewable energy policies in Europe: converging or diverging? Energy Pol 2012;51:192–201. https://doi.org/10.1016/j.enpol.2012.08.064.
[61] Ruusu R, Cao S, Manrique Delgado B, Hasan A. Direct quantification of multiple-source energy flexibility in a residential building using a new model predictive high-level controller. Energy Convers Manag 2019;180:1109–28. https://doi.org/10.1016/j.enconman.2018.11.026.
[62] Verrilli F, Srinivasan S, Gambino G, Canelli M, Himanka M, Del Vecchio C, et al. Model predictive control-based optimal operations of district heating system with thermal energy storage and flexible loads. IEEE Trans Autom Sci Eng 2017;14: 547–57. https://doi.org/10.1109/TASE.2016.2618948.
[63] Verrilli F, Parisio A, Glielmo L. Stochastic model predictive control for optimal energy management of district heating power plants. In: 2016 IEEE 55th conf decis control CDC; 2016 2016. https://doi.org/10.1109/CDC.2016.7798367. 807–12.
[64] Gambino G, Verrilli F, Canelli M, Russo A, Himanka M, Sasso M, et al. Optimal operation of a district heating power plant with thermal energy storage. Proc Am Contr Conf 2016;2016. https://doi.org/10.1109/ACC.2016.7525266. July: 2334–9.

[65] Thombre M, Mdoe Z, Jäschke J. Data-driven robust optimal operation of thermal energy storage in industrial clusters. Processes 2020;8:194. https://doi.org/10.3390/pr8020194.
[66] Candanedo JA, Dehkordi VR, Stylianou M. Model-based predictive control of an ice storage device in a building cooling system. Appl Energy 2013;111:1032–45. https://doi.org/10.1016/j.apenergy.2013.05.081.
[67] Dehkordi VR, Candanedo JA. A model-based predictive control approach for a building cooling system with ice storage. ASME 2013 Dyn Syst Control Conf DSCC 2013 2013:1. https://doi.org/10.1115/DSCC2013-3897.
[68] Beghi A, Cecchinato L, Rampazzo M, Simmini F. Energy efficient control of HVAC systems with ice cold thermal energy storage. J Process Contr 2014;24:773–81. https://doi.org/10.1016/j.jprocont.2014.01.008.
[69] Beghi A, Cecchinato L, Rampazzo M, Simmini F. Modeling and control of HVAC systems with ice cold thermal energy storage. In: 52nd IEEE conf. Decis. Control, IEEE; 2013. p. 4808–13. https://doi.org/10.1109/CDC.2013.6760643.
[70] Pertzborn A. Using distributed agents to optimize thermal energy storage. J Energy Storage 2019;23:89–97. https://doi.org/10.1016/j.est.2019.03.011.
[71] Cao Y, Du J, Soleymanzadeh E. Model predictive control of commercial buildings in demand response programs in the presence of thermal storage. J Clean Prod 2019;218:315–27. https://doi.org/10.1016/j.jclepro.2019.01.266.
[72] Touretzky CR, Salliot AA, Lefevre L, Baldea M. Optimal operation of phase-change thermal energy storage for a commercial building. Proc Am Contr Conf 2015;2015. https://doi.org/10.1109/ACC.2015.7170861. July:980–5.
[73] Touretzky CR, Baldea M. A hierarchical scheduling and control strategy for thermal energy storage systems. Energy Build 2016;110:94–107. https://doi.org/10.1016/j.enbuild.2015.09.049.
[74] Wang J, Zhang Q, Yu Y, Chen X, Yoon S. Application of model-based control strategy to hybrid free cooling system with latent heat thermal energy storage for TBSs. Energy Build 2018;167:89–105. https://doi.org/10.1016/j.enbuild.2018.02.036.
[75] Pangborn HC, Laird CE, Alleyne AG. Hierarchical hybrid MPC for management of distributed phase change thermal energy storage *. In: 2020 Am. Control conf., denver, CO, USA; 2020. p. 4147–53.
[76] Mendoza-Serrano DI, Chmielewski DJ. HVAC control using infinite-horizon economic MPC. Proc IEEE Conf Decis Control 2012. https://doi.org/10.1109/CDC.2012.6426071. 6963–8.
[77] Mendoza-Serrano DI, Chmielewski DJ. Smart grid coordination in building HVAC systems: EMPC and the impact of forecasting. J Process Contr 2014;24:1301–10. https://doi.org/10.1016/j.jprocont.2014.06.005.
[78] Mendoza-Serrano DI, Chmielewski DJ. Smart grid coordination in building hvac systems: computational efficiency of constrained economic linear optimal control. Sci Technol Built Environ 2015;21:812–23. https://doi.org/10.1080/23744731.2015.1057059.
[79] Ma Y, Borrelli F, Hencey B, Coffey B, Bengea S, Haves P. Model predictive control for the operation of building cooling systems. IEEE Trans Contr Syst Technol 2011;20:796–803.
[80] Ma Yudong, Borrelli F, Hencey B, Coffey B, Bengea S, Haves P. Model predictive control for the operation of building cooling systems. IEEE Trans Contr Syst Technol 2012;20:796–803. https://doi.org/10.1109/TCST.2011.2124461.
[81] Ma Y, Borrelli F, Hencey B, Packard A, Bortoff S. Model predictive control of thermal energy storage in building cooling systems. In: Proc. 48h IEEE conf. Decis. Control held jointly with 2009 28th Chinese control conf. IEEE; 2009. p. 392–7.
[82] Berkenkamp F, Gwerder M. Hybrid model predictive control of stratified thermal storages in buildings. Energy Build 2014;84:233–40. https://doi.org/10.1016/j.enbuild.2014.07.052.
[83] Zhao Y, Lu Y, Yan C, Wang S. MPC-based optimal scheduling of grid-connected low energy buildings with thermal energy storages. Energy Build 2015;86:415–26. https://doi.org/10.1016/j.enbuild.2014.10.019.
[84] Pintaldi S, Li J, Sethuvenkatraman S, White S, Rosengarten G. Model predictive control of a high efficiency solar thermal cooling system with thermal storage. Energy Build 2019;196:214–26. https://doi.org/10.1016/j.enbuild.2019.05.008.
[85] Kim SH. Building demand-side control using thermal energy storage under uncertainty: an adaptive Multiple Model-based Predictive Control (MMPC) approach. Build Environ 2013;67:111–28. https://doi.org/10.1016/j.buildenv.2013.05.005.
[86] Touretzky CR, Baldea M. Integrating scheduling and control for economic MPC of buildings with energy storage. J Process Contr 2014;24:1292–300. https://doi.org/10.1016/j.jprocont.2014.04.015.
[87] Kircher KJ, Zhang KM. Model predictive control of thermal storage for demand response. 2015. In: Am. Control conf., chicago, IL, USA; 2015. p. 956–61.
[88] Shan K, Fan C, Wang J. Model predictive control for thermal energy storage assisted large central cooling systems. Energy 2019;179:916–27. https://doi.org/10.1016/j.energy.2019.04.178.
[89] Tang R, Wang S. Model predictive control for thermal energy storage and thermal comfort optimization of building demand response in smart grids. Appl Energy 2019;242:873–82. https://doi.org/10.1016/j.apenergy.2019.03.038.
[90] Yu MG, Pavlak GS. Two-stage stochastic planning for control of building thermal energy storage portfolios with transactive controls. In: 2020 Am. Control conf., denver, CO, USA; 2020. p. 2339–44.
[91] Lee D, Ooka R, Ikeda S, Choi W, Kwak Y. Model predictive control of building energy systems with thermal energy storage in response to occupancy variations and time-variant electricity prices. Energy Build 2020;225. https://doi.org/10.1016/j.enbuild.2020.110291.
[92] Deng K, Sun Y, Chakraborty A, Lu Y, Brouwer J, Mehta PG. Optimal scheduling of chiller plant with thermal energy storage using mixed integer linear programming. In: 2013 Am control conf; 2013. https://doi.org/10.1109/ACC.2013.6580284. 2958–63.
[93] Deng K, Sun Y, Li S, Lu Y, Brouwer J, Mehta PG, et al. Model predictive control of central chiller plant with thermal energy storage via dynamic programming and mixed-integer linear programming. IEEE Trans Autom Sci Eng 2015;12:565–79. https://doi.org/10.1109/TASE.2014.2352280.
[94] Cole WJ, Edgar TF, Novoselac A. Use of model predictive control to enhance the flexibility of thermal energy storage cooling systems. Proc Am Contr Conf 2012:2788–93. https://doi.org/10.1109/acc.2012.6314689.
[95] Bejarano G, Rodríguez D, Lemos JM, Vargas M, Ortega MG. MINLP-based hybrid strategy for operating mode selection of TES-backed-up refrigeration systems. Int J Robust Nonlinear Control 2020;30:6091–111. https://doi.org/10.1002/rnc.4674.
[96] Vasallo MJ, Bravo JM. A MPC approach for optimal generation scheduling in CSP plants. Appl Energy 2016;165:357–70. https://doi.org/10.1016/j.apenergy.2015.12.092.
[97] Vasallo MJ, Bravo JM, Cojocaru EG, Gegúndez ME. Calculating the profits of an economic MPC applied to CSP plants with thermal storage system. Sol Energy 2017;155:1165–77. https://doi.org/10.1016/j.solener.2017.07.033.
[98] Cojocaru EG, Vasallo MJ, Bravo JM, Marin D. A lifetime-extending model-based predictive control for scheduling in concentrating solar power plants. In: 2018 IEEE int. Conf. Ind. Technol. 2018– Febru. IEEE; 2018. p. 1732–7. https://doi.org/10.1109/ICIT.2018.8352444.
[99] Cojocaru EG, Bravo JM, Vasallo MJ, Marin D. Scheduling in concentrating solar power plants based on mixed-integer optimization and binary-regularization. In: 2018 IEEE conf. Decis. Control. 2018– Decem. IEEE; 2018. p. 1632–7. https://doi.org/10.1109/CDC.2018.8619682.
[100] Toub M, Reddy CR, Razmara M, Shahbakhti M, Robinett RD, Aniba G. Model predictive control for MicroCSP integration into a building HVAC system. IEEE Int Conf Control Autom ICCA 2018;2018. https://doi.org/10.1109/ICCA.2018.8444186. June:890–5.
[101] Cojocaru EG, Bravo JM, Vasallo MJ, Marín D. A binary-regularization-based model predictive control applied to generation scheduling in concentrating solar power plants. Optim Contr Appl Methods 2020;41:215–38. https://doi.org/10.1002/oca.2498.
[102] Toub M, Reddy CR, Razmara M, Shahbakhti M, Robinett RD, Aniba G. Model-based predictive control for optimal MicroCSP operation integrated with building HVAC systems. Energy Convers Manag 2019;199:111924. https://doi.org/10.1016/j.enconman.2019.111924.
[103] Liang X, Li Y. Transient analysis and execution-level power tracking control of the concentrating solar thermal power plant. Energies 2019;12:1564. https://doi.org/10.3390/en12081564.
[104] Larsen GKH, van Foreest ND, Scherpen JMA. Distributed MPC applied to a network of households with micro-CHP and heat storage. IEEE Trans Smart Grid 2014;5:2106–14. https://doi.org/10.1109/TSG.2014.2318901.
[105] Fouquet M, Guéguen H, Dumur D, Faille D. Tracking of optimal trajectories for power plants based on physical models. IFAC-PapersOnLine 2015;48:373–8. https://doi.org/10.1016/j.ifacol.2015.12.407.
[106] Wakui T, Kawayoshi H, Yokoyama R, Aki H. Operation management of residential energy-supplying networks based on optimization approaches. Appl Energy 2016;183:340–57. https://doi.org/10.1016/j.apenergy.2016.08.171.
[107] Wu C, Gu W, Jiang P, Li Z, Cai H, Li B. Combined economic dispatch considering the time-delay of district heating network and multi-regional indoor temperature control. IEEE Trans Sustain Energy 2018;9:118–27. https://doi.org/10.1109/TSTE.2017.2718031.
[108] Zhang Y, Meng F, Wang R, Kazemtabrizi B, Shi J. Uncertainty-resistant stochastic MPC approach for optimal operation of CHP microgrid. Energy 2019;179:1265–78. https://doi.org/10.1016/j.energy.2019.04.151.
[109] Horizon 2020 WORK PROGRAMME 2018-2020 general annexes E from P 19-CDC. Technology Readiness Levels Introduction; 2017. p. 7124.